

Asymptotic expansion of the state-density function and the V -filtration

Owen Westfold

Our aim in this note is to recast the asymptotics of the state-density function in the language of $\mathcal{D}$ -module theory, as far as we are able.

Throughout, let X denote a smooth real analytic variety of dimension d , and $f: X \rightarrow \mathbb{R}$ a nonnegative real analytic function.

1 Algebraic foundations

This section is mainly expository, following Popa's notes [4]. On a technical level, our main aim is to verify that the V -filtration construction works in the real analytic setting; due to its connection to monodromy, the V -filtration is normally only considered over $\mathbb{C}$.

1.1 Pushforward along a graph embedding

Let $Y = X \times \mathbb{R}$, with t the coordinate on the second factor, and let $\iota: X \hookrightarrow Y$ denote the graph embedding $x \mapsto (x, f(x))$. We consider the $\mathcal{D}$ -module pushforward $\iota_+ \mathcal{O}_X$. As an $\mathcal{O}_X$ -module, it can be explicitly described as

$$\iota_+ \mathcal{O}_X \cong \mathcal{O}_X \otimes_{\mathbb{R}} \mathbb{R}[\partial_t],$$

with left $\mathcal{D}_Y$ -action given by:

- (1) $g \cdot (h \otimes \partial_t^j) = (gh) \otimes \partial_t^j$, for $g \in \mathcal{O}_X$.
- (2) $t \cdot (h \otimes \partial_t^j) = (fh) \otimes \partial_t^j - jh \otimes \partial_t^{j-1}$.
- (3) $\partial_t \cdot (h \otimes \partial_t^j) = h \otimes \partial_t^{j+1}$.
- (4) $D \cdot (h \otimes \partial_t^j) = D(h) \otimes \partial_t^j - (D(f) \cdot h) \otimes \partial_t^{j+1}$, for $D \in \text{Der}(\mathcal{O}_X)$.

Rule (2) reflects the fact that $t = f$ on the image of ι : if $j = 0$, then $t \cdot (h \otimes 1) = (fh) \otimes 1$, consistent with $t|_{\iota(X)} = f$. The correction term $-jh \otimes \partial_t^{j-1}$ accounts for the commutation relation $[t, \partial_t] = -1$ in $\mathcal{D}_Y$.

There is a useful description of $\iota_+ \mathcal{O}_X$ in terms of the localisation $\mathcal{O}_X[t]_{f-t}$ of $\mathcal{O}_X[t]$ at the element $f-t$. The quotient $\mathcal{O}_X[t]_{f-t}/\mathcal{O}_X[t]$ has a natural $\mathcal{D}_Y$ -module structure, and is generated by the class

$$\delta_f := \left[\frac{1}{f-t} \right],$$

since every element can be written as a finite sum $\sum_{j \geq 0} h_j \partial_t^j \delta_f$ with $h_j \in \mathcal{O}_X$. This shows we have a $\mathcal{D}_Y$ -module isomorphism

$$\iota_+ \mathcal{O}_X \cong \mathcal{O}_X[t]_{f-t}/\mathcal{O}_X[t], \tag{1}$$

which identifies the generator $1 \otimes 1 \in \iota_+ \mathcal{O}_X$ with δ_f .

1.2 The Bernstein-Sato polynomial

The Bernstein–Sato polynomial (or b -function) of f is the key algebraic invariant underlying the V -filtration. To define it, we consider the $\mathcal{D}_X[s]$ -module $\mathcal{O}_X[\frac{1}{f}][s] \cdot f^s$, where f^s is a formal symbol and a derivation $D \in \text{Der}(\mathcal{O}_X)$ acts by

$$D(g \cdot f^s) = \left(D(g) + s g \frac{D(f)}{f} \right) f^s.$$

Theorem 1 (Bernstein–Sato). *There exist a polynomial $b(s) \in \mathbb{R}[s]$ and a differential operator $P(s) \in \mathcal{D}_X[s]$ such that*

$$P(s) \cdot f^{s+1} = b(s) \cdot f^s \quad (2)$$

holds in $\mathcal{O}_X[\frac{1}{f}][s] \cdot f^s$.

Proof. See Björk [2] (Theorem 3.2.6) for a proof in the analytic setting, which covers the real analytic case. $\square$

Definition 2. The *Bernstein–Sato polynomial* $b_f(s)$ is the monic polynomial of minimal degree satisfying (2).

Theorem 3 (Kashiwara). *The roots of $b_f(s)$ are negative rational numbers.*

Example 4. For $f = x^2 + y^2 + z^2$, we have $(\partial_x^2 + \partial_y^2 + \partial_z^2)(x^2 + y^2 + z^2)^{s+1} = (s+1)(4s+6)(x^2 + y^2 + z^2)^s$, so $b_f(s) = (s+1)(s+\frac{3}{2})$. The roots are -1 and $-\frac{3}{2}$.

1.3 The V -filtration

Let $s = -\partial_t t$. The V -filtration $\{V^\gamma\}_{\gamma \in \mathbb{Q}}$ is the unique rational filtration on $\iota_+ \mathcal{O}_X$ which is decreasing, exhaustive, discrete, and left-continuous, and which satisfies the following properties:

- (i) Each V^γ is a coherent module over $\mathcal{D}_X[t, s]$.
- (ii) $t \cdot V^\gamma \subset V^{\gamma+1}$, with equality for $\gamma > 0$.
- (iii) $\partial_t \cdot V^\gamma \subset V^{\gamma-1}$.
- (iv) The operator $s + \gamma$ acts nilpotently on $\text{gr}_V^\gamma := V^\gamma / V^{>\gamma}$, where $V^{>\gamma} := \cup_{\gamma' > \gamma} V^{\gamma'}$.

We record two immediate consequences of the axioms, both used repeatedly below. First, combining (ii) and (iii),

$$s \cdot V^\gamma \subset V^\gamma \quad \text{for every } \gamma, \quad (3)$$

since $s = -\partial_t t$ carries V^γ into $V^{\gamma+1}$ and then back into V^γ . Second, for $\gamma > 0$ the map t induces an isomorphism $\text{gr}_V^\gamma \xrightarrow{\sim} \text{gr}_V^{\gamma+1}$: indeed $V^{\gamma+1} = tV^\gamma$ and $V^{>\gamma+1} = tV^{>\gamma}$ by (ii), and t is injective on $\iota_+ \mathcal{O}_X$ (if $u = \sum_{j \leq J} h_j \otimes \partial_t^j$ with $h_j \neq 0$, then by rule (2) the coefficient of ∂_t^J in tu is $fh_J \neq 0$). In particular the set of positive jumping indices is stable under $\gamma \mapsto \gamma + 1$.

Uniqueness is a formal consequence of (i)–(iv). The proof of existence follows Malgrange’s construction as presented in [4], Theorem 3.2.6. We outline the argument and identify the points that require verification in the real analytic setting.

The original construction, for X a complex algebraic variety, has three steps:

- **Step 1.** One constructs a $\mathcal{D}_X[s]$ -submodule $N' \subset \iota_+ \mathcal{O}_X(*D)$ satisfying $t^k N_f \subset N' \subset t^m N_f$ (where $N_f = \mathcal{D}_X[s]f^s$ is the module generated by f^s), such that $tN' \subset N'$ and the action of $-s$ on N'/tN' has minimal polynomial with roots in $\mathbb{Q} \cap (0, 1]$. This uses the eigenspace decomposition for the action of s on $t^m N_f / t^k N_f$, which is algebraic, and the rationality of the roots of $b_f(s)$ (Kashiwara’s theorem).

- **Step 2.** From N' one defines a filtration W^α on $\iota_+ \mathcal{O}_X(*D)$ by integer shifts: $W^{\beta_i+q} = t^q W^{\beta_i}$ where $\beta_1 < \dots < \beta_\ell$ are the roots of the minimal polynomial from Step 1. The verification that $W^\bullet$ is a V -filtration is algebraic, using the commutation relation $P(s)t = tP(s-1)$ (equation (2.1.2) of [4]).
- **Step 3.** The V -filtration on $\iota_+ \mathcal{O}_X$ is defined by restriction: $V^\alpha \iota_+ \mathcal{O}_X = W^\alpha \cap \iota_+ \mathcal{O}_X$. This uses Lemma 3.2.7 of [4], which shows $N' \subset \iota_+ \mathcal{O}_X$.

Proposition 5. *The V -filtration on $\iota_+ \mathcal{O}_X$ exists and is unique in the real analytic category.*

Proof sketch. The key inputs to the construction are:

- (1) *Existence of the Bernstein–Sato polynomial.* This holds in the real analytic setting by Björk [2].
- (2) *Rationality of the roots of $b_f(s)$.* Since $b_f(s) \in \mathbb{Q}[s]$ (Kashiwara’s theorem, which holds for real analytic functions by the coincidence $b_f = b_{f_\mathbb{C}}$, see [5], §2.5), the roots are rational.
- (3) *Finite length of holonomic modules.* The proof of Step 1 uses the fact that $\mathcal{D}_{X_s} f^s$ is holonomic (over the base field $\mathbb{R}(s)$) and therefore has finite length, to conclude that the chain $\dots \subset \mathcal{D}_{X_s} f^2 f^s \subset \mathcal{D}_{X_s} f f^s \subset \mathcal{D}_{X_s} f^s$ stabilizes. The holonomicity of $\mathcal{D}_{X_s} f^s$ follows from the same argument as in the complex case (the characteristic variety is contained in the zero section of T^*U_s), and the finite length property for holonomic $\mathcal{D}$ -modules holds in the real analytic setting by Björk [2] (Proposition 3.1.5).
- (4) *Algebraic steps.* Steps 2 and 3 of the construction, and the uniqueness proof, are purely algebraic manipulations (eigenspace decompositions, the commutation relation (2.1.2), and the intersection $W^\alpha \cap \iota_+ \mathcal{O}_X$). They apply unchanged.

□

Remark 6. Two points deserve emphasis, since the results of Section 2.3 depend on them.

- (1) Björk’s framework in [2] is complex analytic; the real analytic statements quoted above are obtained by complexifying germs, using that the ring of germs of real analytic functions at $x \in X$ has the ring of germs of holomorphic functions at $x \in X_\mathbb{C}$ as a faithfully flat extension, and that $\mathcal{O}_X$ is coherent in the real analytic category (Cartan). The same complexification underlies the identity $b_f = b_{f_\mathbb{C}}$ of [5], §2.5.
- (2) Axiom (i) — coherence of each V^γ over $\mathcal{D}_X[t, s]$ — is not merely a normalisation. It is used in an essential way in Proposition 27, where local finite generation of V^γ over a compact set is what produces an estimate uniform over all of V^γ . No weaker hypothesis on the filtration suffices there.

A rational γ is a *jumping index* of the V -filtration if $\text{gr}_V^\gamma \neq 0$. Note that the set of jumping indices is *not* bounded below: since $\partial_t \cdot V^\gamma \subset V^{\gamma-1}$ and δ_f generates $\iota_+ \mathcal{O}_X$ over $\mathcal{D}_Y$, the filtration necessarily jumps at arbitrarily negative indices, and in particular $V^0 \subsetneq \iota_+ \mathcal{O}_X$ in general. The indices relevant to the asymptotic expansion are the positive ones, and we write

$$0 < \alpha_1 < \alpha_2 < \dots$$

for the positive jumping indices, and m_k for the nilpotency order of $s + \alpha_k$ on $\text{gr}_V^{\alpha_k}$. By the isomorphism $t: \text{gr}_V^\gamma \xrightarrow{\sim} \text{gr}_V^{\gamma+1}$ noted above, this is an infinite (unbounded) list as soon as it is nonempty, and m_k is constant along each residue class modulo $\mathbb{Z}$.

Corollary 7. *If $\alpha > 0$ is a jumping index of the V -filtration on $\iota_+ \mathcal{O}_X$ (i.e. $\text{gr}_V^\alpha \neq 0$), then α is an integer translate of λ for some root $-\lambda$ of $b_f(s)$.*

Proof. By construction (Step 1), the eigenvalues of $-s$ on N'/tN' are of the form β_i , each congruent to some λ modulo $\mathbb{Z}$ where $-\lambda$ is a root of $b_f(s)$. The V -filtration jumps only at rational numbers of the form $\beta_i + \mathbb{Z}$, giving the result. □

2 Asymptotic analysis of the state density function

In this section, we show that the generator $\delta_f \in \iota_+ \mathcal{O}_X$ has a distributional avatar in the form of the state density function $v(t)$, and we use the V -filtration to control the asymptotic expansion of $v(t)$ as $t \rightarrow 0^+$.

Setup 8. All the statements below are local on X , and in order to have unambiguous formal transposes we fix once and for all a nowhere-vanishing real analytic density dx on X (for instance, the reader may simply assume that X is an open subset of $\mathbb{R}^d$ and that dx is Lebesgue measure). All integrals over X are taken with respect to dx , and the identification of distributions with generalised functions, together with the transpose rules below, is made with respect to dx (and $dx dt$ on Y).

2.1 The distributional solution

Let $C_0^\infty(Y)$ denote the space of test functions on Y , and let $\mathcal{D}'(Y)$ denote its continuous dual — the space of Schwartz distributions on Y . The space $\mathcal{D}'(Y)$ carries a left $\mathcal{D}_Y$ -module structure defined by

$$\langle P \cdot u, \Phi \rangle := \langle u, P^t \Phi \rangle \quad \text{for } P \in \mathcal{D}_Y, u \in \mathcal{D}'(Y), \Phi \in C_0^\infty(Y),$$

where P^t is the formal transpose: $g^t = g$ for multiplication operators, $\partial_{x_i}^t = -\partial_{x_i}$, $\partial_t^t = -\partial_t$, and $(PQ)^t = Q^t P^t$.

Definition 9. The distribution $\Delta_f \in \mathcal{D}'(Y)$ is defined by

$$\langle \Delta_f, \Phi \rangle := \int_X \Phi(x, f(x)) dx \quad \text{for } \Phi \in C_0^\infty(Y).$$

In the usual notation, $\Delta_f = \delta(t - f(x))$: it is supported on the graph $\Gamma_f = \{(x, t) : t = f(x)\}$ and integrates test functions over it.

Proposition 10. *There is a unique $\mathcal{D}_Y$ -module homomorphism*

$$\sigma: \iota_+ \mathcal{O}_X \longrightarrow \mathcal{D}'(Y)$$

satisfying $\sigma(\delta_f) = \Delta_f$. It is given by $\sigma(h \partial_t^j \delta_f) = h(x) \cdot \partial_t^j \Delta_f$ for all $h \in \mathcal{O}_X$ and $j \geq 0$.

Proof. Uniqueness. Since δ_f generates $\iota_+ \mathcal{O}_X$ as a $\mathcal{D}_Y$ -module (Section 1.1), any $\mathcal{D}_Y$ -homomorphism is determined by $\sigma(\delta_f)$. The formula $\sigma(h \partial_t^j \delta_f) = h \partial_t^j \Delta_f$ then follows from the $\mathcal{D}_Y$ -linearity and the action rules (1)–(4).

Existence. We must verify that Δ_f satisfies the defining relations of $\iota_+ \mathcal{O}_X$, i.e., that σ respects the $\mathcal{D}_Y$ -action. Rules (1) and (3) (action of $g \in \mathcal{O}_X$ and ∂_t) are immediate from the definitions. The two nontrivial relations are:

Action of t (Rule 2). We must show $t \cdot \partial_t^j \Delta_f = f \partial_t^j \Delta_f - j \partial_t^{j-1} \Delta_f$. Pairing the left side with $\Phi \in C_0^\infty(Y)$:

$$\langle t \cdot \partial_t^j \Delta_f, \Phi \rangle = (-1)^j \int_X \partial_t^j (t \Phi) \big|_{t=f(x)} dx.$$

By the Leibniz rule, $\partial_t^j (t \Phi) = t \partial_t^j \Phi + j \partial_t^{j-1} \Phi$. Evaluating at $t = f(x)$ and comparing with the right side (where $f(x)$ and $\partial_{x_i} f$ are independent of t , so pass through ∂_t^j) gives the result.

Action of ∂_{x_i} (Rule 4). We must show $\partial_{x_i} \cdot \partial_t^j \Delta_f = -(\partial_{x_i} f) \partial_t^{j+1} \Delta_f$. Pairing the left side with Φ :

$$\langle \partial_{x_i} \partial_t^j \Delta_f, \Phi \rangle = (-1)^{j+1} \int_X (\partial_t^j \partial_{x_i} \Phi)(x, f(x)) dx.$$

Pairing the right side:

$$\langle -(\partial_{x_i} f) \partial_t^{j+1} \Delta_f, \Phi \rangle = (-1)^j \int_X (\partial_{x_i} f)(x) (\partial_t^{j+1} \Phi)(x, f(x)) dx.$$

These are equal by the chain rule identity

$$\frac{\partial}{\partial x_i} [(\partial_t^j \Phi)(x, f(x))] = (\partial_{x_i} \partial_t^j \Phi)(x, f(x)) + (\partial_{x_i} f)(x) (\partial_t^{j+1} \Phi)(x, f(x)),$$

integrated over X : the left side vanishes (since Φ has compact support, the composite $x \mapsto (\partial_t^j \Phi)(x, f(x))$ is compactly supported, and the integral of a total derivative of a compactly supported function is zero), giving the required identity. $\square$

Remark 11. Rules (1)–(4) describe the action of the subring of $\mathcal{D}_Y$ generated by $\mathcal{O}_X$, $\text{Der}(\mathcal{O}_X)$, t and ∂_t , whereas $\mathcal{O}_Y$ also contains functions that are genuinely analytic in t . This is harmless: for $g \in \mathcal{O}_Y$ one checks directly from Definition 9 that

$$g \cdot \partial_t^j \Delta_f = \sum_{i=0}^j (-1)^i \binom{j}{i} (\partial_t^i g)(x, f(x)) \partial_t^{j-i} \Delta_f,$$

so that the action of $\mathcal{O}_Y$ is determined by the listed generators together with the relation $t \mapsto f$, and it suffices to verify the relations (1)–(4).

2.2 The state density function

Note that pairing Δ_f with a test function of the form $\Phi(x, t) = \phi(x)\psi(t)$ gives $\int_X \phi(x)\psi(f(x)) dx$. This leads to the following definition.

Definition 12. Let $\phi \in C_0^\infty(X)$. The *state density distribution* associated to f and ϕ is the distribution $v_\phi \in \mathcal{D}'(\mathbb{R})$ defined by

$$\int_{\mathbb{R}} \psi(t) v_\phi(t) dt := \int_X \psi(f(x)) \phi(x) dx \quad \text{for all } \psi \in C_0^\infty(\mathbb{R}). \quad (4)$$

Equivalently, v_ϕ is the pushforward of the measure $\phi(x) dx$ under $f: X \rightarrow \mathbb{R}$. It is a compactly supported distribution on $[0, \infty)$, and it is a positive measure when $\phi \geq 0$. We abbreviate $v := v_\phi$ when ϕ is fixed.

We write $v_\phi(t) = \int_X \delta(t - f(x)) \phi(x) dx$ as a formal shorthand for (4).

Remark 13. The positivity hypothesis $\phi \geq 0$ is *not* imposed in Definition 12. This matters: even when the test function of interest is nonnegative, the analysis below forces us to consider $v_{h\phi}$ for arbitrary $h \in \mathcal{O}_X$ (Proposition 17). Positivity of ϕ will only be used where it is genuinely needed, namely in the two-sided bounds of Section ??.

Proposition 14. *The distribution v_ϕ is represented by a locally integrable function on $(0, \infty)$, smooth away from the critical values of $f|_{\text{supp } \phi}$. At a regular value $t_0 > 0$ of f , it is given by the coarea formula:*

$$v_\phi(t_0) = \int_{f^{-1}(t_0)} \frac{\phi(x)}{|\nabla f(x)|} dS,$$

where dS is the surface measure on the level set $f^{-1}(t_0)$.

Proof. See Watanabe [6], Theorem 4.3. $\square$

The function $v_\phi(t)$ is called the *state density function*.

Lemma 15. *Fix a compact $K \subset X$. The set of critical values of $f|_K$ is finite. In particular there exists $\epsilon_K > 0$ such that $f|_K$ has no critical value in $(0, \epsilon_K)$, and consequently v_ϕ is smooth on $(0, \epsilon_K)$ for every $\phi \in C_0^\infty(X)$ with $\text{supp } \phi \subset K$.*

Proof. The critical set $C = \{x \in X : df_x = 0\}$ is a closed real analytic subset, so $C \cap K$ is compact and subanalytic; by the analytic Sard theorem $f(C)$ has measure zero, and $f(C \cap K)$ is a compact subanalytic subset of $\mathbb{R}$, hence a finite union of points and closed intervals. Having measure zero, it is finite. $\square$

Remark 16. Some hypothesis of this kind is necessary: on a noncompact X the set of critical values of a real analytic function need not be discrete. For example $f(x) = x + x^2 \sin(1/x)$ is real analytic on $(0, 1)$ and has critical values accumulating at 0. Since v_ϕ depends only on $f|_{\text{supp } \phi}$, restricting to a compact set is no loss.

For any $u \in \iota_+ \mathcal{O}_X$ defined on a neighbourhood of $\text{supp } \phi$, and any $\phi \in C_0^\infty(X)$, define the distribution $\langle u, \phi \rangle \in \mathcal{D}'(\mathbb{R})$ by

$$\int_{\mathbb{R}} \psi(t) \langle u, \phi \rangle(t) dt := \langle \sigma(u), \phi(x)\psi(t) \rangle \quad \text{for all } \psi \in C_0^\infty(\mathbb{R}).$$

In particular, $\langle \delta_f, \phi \rangle = v_\phi$.

Proposition 17. *Let $K \subset X$ be compact and ϵ_K as in Lemma 15. For any $\phi \in C_0^\infty(X)$ with $\text{supp } \phi \subset K$ and any $u \in \iota_+ \mathcal{O}_X$ defined near K , the distribution $\langle u, \phi \rangle$ is represented by a smooth function on $(0, \epsilon_K)$. Explicitly, if $u = \sum_j h_j \partial_t^j \delta_f$ (a finite sum, $h_j \in \mathcal{O}_X$), then*

$$\langle u, \phi \rangle = \sum_j (-1)^j \partial_t^j v_{h_j \phi}. \quad (5)$$

Proof. By Proposition 10, $\sigma(u) = \sum_j h_j \partial_t^j \Delta_f$. Pairing with $\phi(x)\psi(t)$ and moving ∂_t^j onto ψ (picking up the sign $(-1)^j$) gives (5), where $v_{h_j \phi}$ is the state density with test function $h_j \phi \in C_0^\infty(X)$, supported in K . By Lemma 15 each $v_{h_j \phi}$ is smooth on $(0, \epsilon_K)$, hence so is each derivative, and the sum is finite. $\square$

Remark 18. Equation (5) corrects a sign that is immaterial for the estimates below but matters for the bookkeeping of Lemma 19.

The $\mathcal{D}_Y$ -linearity of σ now translates into the following intertwining rules, which are the bridge between the algebra of Section 1 and the analysis of Section 2.3.

Lemma 19. *Let $\phi \in C_0^\infty(X)$ and let $u \in \iota_+ \mathcal{O}_X$ be defined near $\text{supp } \phi$. Then, as distributions on $\mathbb{R}$:*

- (a) $\langle t \cdot u, \phi \rangle = t \cdot \langle u, \phi \rangle$;
- (b) $\langle s \cdot u, \phi \rangle = (-\partial_t t) \cdot \langle u, \phi \rangle$, where $s = -\partial_t t$;
- (c) $\langle P \cdot u, \phi \rangle = \langle u, P^t \phi \rangle$ for every $P \in \mathcal{D}_X$, where P^t is the formal transpose on X . In particular $\text{supp}(P^t \phi) \subset \text{supp } \phi$.

Proof. For (a): $\langle \sigma(tu), \phi \psi \rangle = \langle t \sigma(u), \phi \psi \rangle = \langle \sigma(u), \phi(t\psi) \rangle$, which is the pairing of $t \cdot \langle u, \phi \rangle$ with ψ . For (b): the transpose of $-\partial_t t$ is $t \partial_t$, so $\langle \sigma(su), \phi \psi \rangle = \langle \sigma(u), \phi \cdot t \partial_t \psi \rangle$, which is the pairing of $(-\partial_t t) \langle u, \phi \rangle$ with ψ . For (c): P involves only x and ∂_x , so $P^t(\phi(x)\psi(t)) = (P^t \phi)(x) \psi(t)$ and $\langle \sigma(Pu), \phi \psi \rangle = \langle \sigma(u), (P^t \phi) \psi \rangle$. $\square$

Note the asymmetry in Lemma 19: t and s act on the variable t of the pairing, whereas the operators of $\mathcal{D}_X$ are transposed onto the test function ϕ , *without enlarging its support*. This is the mechanism that will make our estimates uniform.

2.3 The asymptotic scale

Definition 20. A *Nilsson symbol* is a formal series

$$S = \sum_{\gamma \in E} \sum_{j=0}^J c_{\gamma j} t^{\gamma-1} (-\log t)^j, \quad c_{\gamma j} \in \mathbb{R},$$

where $J \in \mathbb{Z}_{\geq 0}$ and $E \subset \mathbb{Q}$ is discrete and bounded below. We call the γ occurring with $c_{\gamma j} \neq 0$ the *exponents* of S and set

$$\text{ord}(S) := \min\{\gamma : c_{\gamma j} \neq 0 \text{ for some } j\} \in \mathbb{Q} \cup \{+\infty\},$$

with $\text{ord}(0) = +\infty$. For $\gamma \in E$, the γ -*block* of S is the finite sum $\sum_{j \leq J} c_{\gamma j} t^{\gamma-1} (-\log t)^j$.

Definition 21. Let w be a smooth function on $(0, \epsilon)$. We say w has *asymptotic symbol* S , and write $\tau(w) = S$, if for every $A \in \mathbb{R}$

$$w(t) - \sum_{\gamma \in E, \gamma \leq A} \sum_{j \leq J} c_{\gamma j} t^{\gamma-1} (-\log t)^j = O(t^A) \quad (t \rightarrow 0^+).$$

We write $\mathcal{N}(\epsilon)$ for the set of smooth w on $(0, \epsilon)$ which have an asymptotic symbol and for which the symbol may be differentiated term by term to all orders, i.e. $\partial_t^\ell w \in \mathcal{N}(\epsilon)$ with $\tau(\partial_t^\ell w) = \partial_t^\ell \tau(w)$ for all $\ell \geq 0$.

Since the functions $t^{\gamma-1} (-\log t)^j$ form an asymptotic scale (they are totally ordered by decay as $t \rightarrow 0^+$), the symbol of w is unique when it exists. We record the elementary properties we shall use.

Lemma 22. $\mathcal{N}(\epsilon)$ is an $\mathbb{R}$ -vector space, stable under ∂_t and under multiplication by t , and τ is linear with

$$\tau(\partial_t w) = \partial_t \tau(w), \quad \tau(t w) = t \tau(w).$$

Consequently $\mathcal{N}(\epsilon)$ is stable under $s = -\partial_t$, and s acts on symbols block by block:

$$s \cdot (t^{\gamma-1} (-\log t)^j) = -\gamma t^{\gamma-1} (-\log t)^j + j t^{\gamma-1} (-\log t)^{j-1}. \quad (6)$$

Moreover, if $w \in \mathcal{N}(\epsilon)$ and $\text{ord}(\tau(w)) = \gamma_0$ with top logarithmic degree j_0 , then $w(t) = O(t^{\gamma_0-1} |\log t|^{j_0})$; and if all exponents of $\tau(w)$ are $\geq \alpha$ then $w(t) = O(t^{\alpha-1} |\log t|^J)$ for J the top logarithmic degree of $\tau(w)$.

Proof. All assertions are immediate from Definition 21, except (6), which is the computation $-\partial_t(t^\gamma (-\log t)^j) = -\gamma t^{\gamma-1} (-\log t)^j + j t^{\gamma-1} (-\log t)^{j-1}$. Note that $t \cdot$ raises exponents by 1, ∂_t lowers them by 1, and s preserves each γ -block (acting there as $-\gamma \text{Id} + N$, with N the nilpotent operator lowering the logarithmic degree). $\square$

2.4 The transcendental input

Axioms (i)–(iv) of the V -filtration are purely formal, and no purely formal argument can produce a growth estimate for $\langle u, \phi \rangle$: the same axioms are satisfied by V -filtrations on irregular modules, whose distributional solutions may contain flat contributions such as $e^{-1/t}$ that no algebraic manipulation can detect. Any proof of an estimate such as Theorem 30 must therefore consume some transcendental input. We take as that input the existence (but *not* the shape) of an asymptotic expansion for the state density, which is a theorem of Jeanquartier [7] obtained from Hironaka's resolution of singularities. The role of the V -filtration is then to *locate* the exponents and to bound the logarithmic degrees. This is the same division of labour as in the complex case, where Malgrange's theorem on asymptotic integrals [8] plays the role of Theorem 23.

Theorem 23 (Jeanquartier). *Let $K \subset X$ be compact. There exist $\epsilon_K > 0$, a finite set $A_K \subset \mathbb{Q}_{>0}$ and an integer $J_K \geq 0$ with the following property. Set $E_K := A_K + \mathbb{Z}_{\geq 0}$. For every $\psi \in C_0^\infty(X)$ with $\text{supp } \psi \subset K$, the state density v_ψ lies in $\mathcal{N}(\epsilon_K)$, and its symbol has the form*

$$\tau(v_\psi) = \sum_{\gamma \in E_K} \sum_{j=0}^{J_K} c_{\gamma j}(\psi) t^{\gamma-1} (-\log t)^j.$$

In particular the exponents and logarithmic degrees occurring in $\tau(v_\psi)$ are bounded by data depending only on K , not on ψ ; and all exponents are strictly positive.

Proof. This is Jeanquartier [7]; see also Barlet [9], [10, §II.7] and Watanabe [6, Theorem 4.6]. We recall the shape of the argument, since it is what makes the uniformity in ψ visible. Choose (Hironaka [3]) a proper real analytic map $\pi: X' \rightarrow X$, an isomorphism away from $f^{-1}(0)$, such that on each chart of a finite atlas covering the compact set $\pi^{-1}(K)$ one has

$$f \circ \pi = u(x') \prod_i x_i'^{m_i}, \quad |\det d\pi| = |u'(x')| \prod_i x_i'^{a_i}$$

with u, u' nowhere vanishing. Pulling back and using a partition of unity subordinate to the atlas writes v_ψ as a finite sum of model integrals $\int \delta(t - u \prod_i x_i'^{m_i}) \prod_i x_i'^{a_i} \Psi_\ell(x') dx'$ with $\Psi_\ell \in C_0^\infty$. Each model integral has an asymptotic expansion in the scale, with exponents in $\{(a_i + 1)/m_i\} + \mathbb{Z}_{\geq 0}$ and logarithmic degree at most (number of i with $m_i \neq 0$) $-1 \leq d - 1$, and the expansion is differentiable term by term. This gives the statement with

$$A_K = \left\{ \frac{a_i + 1}{m_i} \right\}, \quad J_K \leq d - 1,$$

the union being over the finitely many divisorial data occurring over K ; these depend on K but not on ψ . Positivity of the exponents is the statement $a_i + 1 > 0$, which holds since $a_i \geq 0$. $\square$

Lemma 24. *Let $K \subset X$ be compact and $\phi \in C_0^\infty(X)$ with $\text{supp } \phi \subset K$. Let $u \in \iota_+ \mathcal{O}_X$ be defined near K , and write $u = \sum_{j \leq J} h_j \partial_t^j \delta_f$. Then $\langle u, \phi \rangle \in \mathcal{N}(\epsilon_K)$, and the exponents of $\tau(\langle u, \phi \rangle)$ lie in $E_K - J$ while its logarithmic degrees are at most J_K . In particular*

$$\text{ord } \tau(\langle u, \phi \rangle) \geq \epsilon_K^0 - J, \quad \epsilon_K^0 := \min A_K > 0.$$

Proof. Immediate from (5), Theorem 23 and Lemma 22: each $h_j \phi$ is a test function supported in K , so $v_{h_j \phi} \in \mathcal{N}(\epsilon_K)$ with exponents in E_K ; applying ∂_t^j lowers exponents by j and does not raise logarithmic degrees; and $\mathcal{N}(\epsilon_K)$ is a vector space. $\square$

2.5 The uniform seed

The estimate of Lemma 24 degrades as the ∂_t -degree of u grows, and V^γ contains elements of arbitrarily large ∂_t -degree (for instance $t^k \partial_t^k \delta_f \in V^\gamma$ whenever $\delta_f \in V^\gamma$, since $t^k \partial_t^k$ is a polynomial in $t \partial_t = -(s + 1)$). So an element-by-element application of Lemma 24 yields no bound valid on all of V^γ . The missing uniformity comes from coherence — axiom (i) — which we now put to work. It is the only place where axiom (i) is used, and it is what replaces the (incorrect) a priori bound of the first version of this note.

Definition 25. For $K \subset X$ compact and $u \in \iota_+ \mathcal{O}_X$ defined near K , set

$$\omega_K(u) := \inf \{ \text{ord } \tau(\langle u, \psi \rangle) : \psi \in C_0^\infty(X), \text{supp } \psi \subset K \} \in \mathbb{Q} \cup \{+\infty\}.$$

Lemma 26. *Let $K \subset X$ be compact.*

$$(i) \quad \omega_K(u) \geq \epsilon_K^0 - \deg_{\partial_t}(u) > -\infty.$$

(ii) $\omega_K(u + u') \geq \min(\omega_K(u), \omega_K(u'))$, and $\omega_{K'}(u) \geq \omega_K(u)$ whenever $K' \subset K$.

(iii) $\omega_K(P \cdot u) \geq \omega_K(u)$ for every $P \in \mathcal{D}_X[t, s]$ defined near K . Moreover $\omega_K(t^n u) = n + \omega_K(u)$ for $n \geq 0$.

Proof. (i) is Lemma 24, and (ii) is clear. For (iii), it suffices by additivity to treat P a word in the generators $\mathcal{O}_X$, $\text{Der}(\mathcal{O}_X)$, t , s of the ring $\mathcal{D}_X[t, s]$, and we induct on the length of the word. If $P = Q P'$ with Q one of the generators, then by Lemma 19:

- if $Q \in \mathcal{D}_X$, then $\langle Pu, \psi \rangle = \langle P'u, Q^t \psi \rangle$ and $\text{supp}(Q^t \psi) \subset \text{supp} \psi \subset K$, so the induction hypothesis applies to P' with the test function $Q^t \psi$;
- if $Q = t$, then $\tau(\langle Pu, \psi \rangle) = t \cdot \tau(\langle P'u, \psi \rangle)$, which raises ord by exactly 1;
- if $Q = s$, then $\tau(\langle Pu, \psi \rangle) = s \cdot \tau(\langle P'u, \psi \rangle)$, and by (6) the operator s preserves each γ -block, hence does not lower ord.

In all three cases $\text{ord } \tau(\langle Pu, \psi \rangle) \geq \omega_K(P'u) \geq \omega_K(u)$. The last assertion follows since $\text{ord } \tau(\langle t^n u, \psi \rangle) = n + \text{ord } \tau(\langle u, \psi \rangle)$ for every ψ . $\square$

Proposition 27 (Uniform seed). *Let $K \subset X$ be compact. There is a constant $C_K \geq 0$ such that*

$$\omega_K(u) \geq \gamma - 1 - C_K \quad \text{for every rational } \gamma > 0 \text{ and every } u \in V^\gamma \text{ defined near } K.$$

Proof. Step 1: the strip $\beta \in (0, 1]$. Since the V -filtration is discrete, the family $\{V^\beta : \beta \in (0, 1]\}$ consists of finitely many distinct submodules; fix one, $V := V^\beta$.

By axiom (i), V is a coherent $\mathcal{D}_X[t, s]$ -module, hence locally finitely generated: every $y \in K$ has an open neighbourhood B_y , which we may take with $\overline{B_y}$ compact, and finitely many sections $u^{(1, y)}, \dots, u^{(p_y, y)} \in V(B_y)$ whose germs generate the stalk V_z over $\mathcal{D}_X[t, s]_z$ for every $z \in B_y$. By compactness, choose $y_1, \dots, y_M$ with $K \subset \bigcup_\nu B_{y_\nu}$, and set

$$-C_\beta := \min_{1 \leq \nu \leq M} \min_{1 \leq i \leq p_{y_\nu}} \omega_{\overline{B_{y_\nu}}}(u^{(i, \nu)}),$$

which is finite by Lemma 26(i). Crucially, C_β depends only on K and on the chosen generators — not on the element u we are about to estimate.

Now let $u \in V$ be defined near K , and let $\psi \in C_0^\infty(X)$ with $\text{supp} \psi \subset K$. For each $z \in \text{supp} \psi$ choose $\nu(z)$ with $z \in B_{y_{\nu(z)}}$. Since the germs of $u^{(1, \nu(z))}, \dots$ generate V_z , we may write $u = \sum_i P_i^z u^{(i, \nu(z))}$ with $P_i^z \in \mathcal{D}_X[t, s]$ on some open $U_z \ni z$ with $\overline{U_z} \subset B_{y_{\nu(z)}}$ compact. Cover the compact set $\text{supp} \psi$ by finitely many $U_{z_1}, \dots, U_{z_R}$ and choose a partition of unity, writing $\psi = \sum_\rho \psi_\rho$ with $\text{supp} \psi_\rho \subset \overline{U_{z_\rho}}$.

The pairing $\langle \cdot, \psi_\rho \rangle$ depends only on the restriction of its first argument to a neighbourhood of $\text{supp} \psi_\rho$ (by (5)), so

$$\langle u, \psi_\rho \rangle = \left\langle \sum_i P_i^{z_\rho} u^{(i, \nu(z_\rho))}, \psi_\rho \right\rangle.$$

Applying Lemma 26(ii),(iii) on the compact set $\overline{U_{z_\rho}}$,

$$\text{ord } \tau(\langle u, \psi_\rho \rangle) \geq \min_i \omega_{\overline{U_{z_\rho}}}(u^{(i, \nu(z_\rho))}) \geq \min_i \omega_{\overline{B_{y_{\nu(z_\rho)}}}}(u^{(i, \nu(z_\rho))}) \geq -C_\beta,$$

using $\overline{U_{z_\rho}} \subset \overline{B_{y_{\nu(z_\rho)}}}$ and monotonicity of ω in K . Summing over ρ gives $\text{ord } \tau(\langle u, \psi \rangle) \geq -C_\beta$, and taking the infimum over ψ , $\omega_K(u) \geq -C_\beta$.

Step 2: integer shifts. Put $C_K := \max\{C_\beta : \beta \in (0, 1]\}$ (a maximum over finitely many values). Let $\gamma > 0$ and $u \in V^\gamma$. Write $\gamma = \beta + n$ with $n = \lceil \gamma \rceil - 1 \in \mathbb{Z}_{\geq 0}$ and $\beta = \gamma - n \in (0, 1]$. By axiom (ii), $V^\gamma = t^n V^\beta$, so $u = t^n u'$ with $u' \in V^\beta$. By Lemma 26(iii) and Step 1,

$$\omega_K(u) = n + \omega_K(u') \geq n - C_K \geq \gamma - 1 - C_K. \quad \square$$

Remark 28. Proposition 27 is already a growth statement — pairings of deep elements of the filtration decay fast — but with a lossy constant C_K that has nothing to do with the geometry of f . The point of the next theorem is that the V -filtration can be used to bootstrap this crude bound into the sharp one, by comparison rather than by an ODE argument.

2.6 Regularity along the V -filtration

We first record the purely algebraic fact that a suitable polynomial in s pushes an element arbitrarily far up the filtration. Recall that $s \cdot V^\gamma \subset V^\gamma$ for all γ , by axioms (ii) and (iii).

Lemma 29. *Let $\gamma_* \leq B$ be rational, and let α_k run over the jumping indices of the V -filtration lying in $[\gamma_*, B]$, with nilpotency orders m_k . Put*

$$Q_{\gamma_*, B}(s) := \prod_{\gamma_* \leq \alpha_k \leq B} (s + \alpha_k)^{m_k} \in \mathbb{Q}[s]$$

(the empty product being 1). Then $Q_{\gamma_*, B}(s) \cdot V^{\gamma_*} \subset V^{>B}$.

Proof. Induction on the number r of jumping indices in $[\gamma_*, B]$. If $r = 0$ then, by discreteness and left-continuity, V^γ is constant for $\gamma \in [\gamma_*, B]$ and also for γ slightly larger than B , whence $V^{\gamma_*} = V^{>B}$ and there is nothing to prove. If $r \geq 1$, let α_{k_0} be the least jumping index in $[\gamma_*, B]$; by left-continuity $V^{\gamma_*} = V^{\alpha_{k_0}}$, and axiom (iv) gives $(s + \alpha_{k_0})^{m_{k_0}} V^{\alpha_{k_0}} \subset V^{>\alpha_{k_0}}$. Now $V^{>\alpha_{k_0}} = V^{\gamma'_*}$ for some $\gamma'_* > \alpha_{k_0}$ (discreteness), the interval $[\gamma'_*, B]$ contains $r - 1$ jumping indices, and the remaining factors of $Q_{\gamma_*, B}$ are exactly $Q_{\gamma'_*, B}$. Since the factors of $Q_{\gamma_*, B}(s)$ commute and s preserves each V^γ , we may apply them in this order and conclude by induction. $\square$

Theorem 30 (Regularity along the V -filtration). *Let $\alpha > 0$ be rational and let $u \in V^\alpha$. Let $\phi \in C_0^\infty(X)$ with $K := \text{supp } \phi$, and suppose u is defined near K . Then every exponent of $\tau(\langle u, \phi \rangle)$ is $\geq \alpha$. Consequently*

$$\langle u, \phi \rangle = O(t^{\alpha-1} |\log t|^N) \quad (t \rightarrow 0^+),$$

where $N := J_K$ may be taken to depend only on $\text{supp } \phi$ — not on u or on α .

Proof. Write $w := \langle u, \phi \rangle \in \mathcal{N}(\epsilon_K)$ (Lemma 24) and suppose, for contradiction, that $\tau(w)$ has a nonzero coefficient at some exponent $< \alpha$. The exponents of $\tau(w)$ form a discrete set bounded below, so there is a least one, $\gamma_0 := \text{ord } \tau(w) < \alpha$.

Fix a rational $B > \alpha$ and set $Q_B(s) := Q_{\alpha, B}(s)$ as in Lemma 29, so that

$$Q_B(s) \cdot u \in V^{>B} \subset V^\gamma \quad \text{for some } \gamma > B.$$

Since $\gamma > B > \alpha > 0$, Proposition 27 applies and gives

$$\text{ord } \tau(\langle Q_B(s)u, \phi \rangle) \geq \omega_K(Q_B(s)u) \geq \gamma - 1 - C_K > B - 1 - C_K. \quad (7)$$

On the other hand, by Lemma 19(b) and Lemma 22,

$$\tau(\langle Q_B(s)u, \phi \rangle) = Q_B(s) \cdot \tau(w),$$

where s acts on symbols by (6). On the γ_0 -block — the finite dimensional space spanned by $t^{\gamma_0-1}(-\log t)^j$, $0 \leq j \leq J_K$ — the operator s acts as $-\gamma_0 \text{Id} + N$ with N nilpotent, so each factor $s + \alpha_k$ acts as $(\alpha_k - \gamma_0) \text{Id} + N$. Every jumping index occurring in Q_B satisfies $\alpha_k \geq \alpha > \gamma_0$, so each such factor is invertible on the γ_0 -block, and hence so is $Q_B(s)$. Therefore $Q_B(s)\tau(w)$ still has a nonzero γ_0 -block; since s does not lower exponents, no lower exponent is created, and

$$\text{ord } \tau(\langle Q_B(s)u, \phi \rangle) = \gamma_0.$$

Comparing with (7) gives $\gamma_0 > B - 1 - C_K$ for every $B > \alpha$, which is absurd as soon as $B > \gamma_0 + 1 + C_K$.

Hence all exponents of $\tau(w)$ are $\geq \alpha$, and the growth estimate follows from Lemma 22. $\square$

Remark 31. The proof uses each axiom exactly once, and it is worth isolating where: (i) coherence gives the uniform seed (Proposition 27); (ii) $t \cdot V^\gamma = V^{\gamma+1}$ reduces the seed to the strip $(0, 1]$; (iii) enters through $s \cdot V^\gamma \subset V^\gamma$; and (iv) supplies the polynomial $Q_B(s)$ that pushes u up the filtration. The transcendental content is confined to Theorem 23.

2.7 The asymptotic expansion

Theorem 32. *Let $\phi \in C_0^\infty(X)$. The state density function $v_\phi = \langle \delta_f, \phi \rangle$ admits an asymptotic expansion as $t \rightarrow 0^+$ of the form*

$$v_\phi(t) \sim \sum_{k \geq 1} \sum_{m=1}^{m_k} D_{km}(\phi) t^{\alpha_k-1} (-\log t)^{m-1},$$

where $0 < \alpha_1 < \alpha_2 < \dots$ are the positive jumping indices of the V -filtration on $\iota_+ \mathcal{O}_X$ and m_k is the nilpotency order of $s + \alpha_k$ on $\mathrm{gr}_V^{\alpha_k}$.

Proof. By Theorem 23, $v_\phi \in \mathcal{N}(\epsilon_K)$ ($K = \mathrm{supp} \phi$) and every exponent of $\tau(v_\phi)$ is positive. It therefore suffices to prove:

- (a) every exponent γ of $\tau(v_\phi)$ is a jumping index of the V -filtration; and
- (b) if $\gamma = \alpha_k$ is such an exponent, then its logarithmic degree is at most $m_k - 1$.

Since the filtration is exhaustive and discrete, there is a largest $\gamma_* \in \mathbb{Q}$ with $\delta_f \in V^{\gamma_*}$. (We do *not* need to know that $\gamma_* > 0$; in the complex case $\gamma_* = \mathrm{lct}(f)$, but we shall not use this.)

Let $\gamma > 0$ be an exponent of $\tau(v_\phi)$. First, $\gamma \geq \gamma_*$: this is vacuous if $\gamma_* \leq 0$, and otherwise it follows from Theorem 30 applied to $\delta_f \in V^{\gamma_*}$. Now set

$$Q(s) := Q_{\gamma_*, \gamma}(s) = \prod_{\gamma_* \leq \alpha_k \leq \gamma} (s + \alpha_k)^{m_k},$$

the product being over *all* jumping indices in $[\gamma_*, \gamma]$ (of either sign). By Lemma 29, $u := Q(s) \delta_f \in V^{>\gamma}$, so $u \in V^{\gamma'}$ for some $\gamma' > \gamma > 0$, and Theorem 30 gives

$$\text{every exponent of } \tau(\langle u, \phi \rangle) = Q(s) \tau(v_\phi) \text{ is } \geq \gamma' > \gamma. \quad (8)$$

In particular the γ -block of $Q(s) \tau(v_\phi)$ vanishes.

Now examine the action of $Q(s)$ on the γ -block of $\tau(v_\phi)$. As in the proof of Theorem 30, $s + \alpha_k$ acts there as $(\alpha_k - \gamma) \mathrm{Id} + N$ with N nilpotent (lowering the logarithmic degree, $N(t^{\gamma-1}(-\log t)^j) = j t^{\gamma-1}(-\log t)^{j-1}$). Every factor with $\alpha_k \neq \gamma$ is therefore invertible on this block.

If γ is not a jumping index, all factors are invertible, so $Q(s)$ is invertible on the γ -block; by (8) that block must vanish, contradicting the assumption that γ is an exponent of $\tau(v_\phi)$. This proves (a).

If $\gamma = \alpha_k$, then $Q(s)$ acts on the block as an invertible operator composed with N^{m_k} . Since N^{m_k} annihilates exactly the terms of logarithmic degree $\leq m_k - 1$ and is injective on the span of the terms of degree $\geq m_k$, (8) forces all coefficients of logarithmic degree $\geq m_k$ to vanish. This proves (b), and hence the theorem, with $D_{km}(\phi) := c_{\alpha_k, m-1}(\phi)$. $\square$

Remark 33. Theorem 32 does not assert that $D_{km}(\phi) \neq 0$; the coefficient at a jumping index may well vanish for every ϕ . Section 3 shows that this really happens, and that the discrepancy is measured by the difference between the (complex) log canonical threshold and the real one.

Lemma 34. *For each positive jumping index α_k and each $m = 1, \dots, m_k$, the map $\phi \mapsto D_{km}(\phi)$ is a distribution on X of finite order.*

Proof. Linearity is clear: $\phi \mapsto v_\phi$ is linear, and the asymptotic symbol of a function is unique, so the coefficients $c_{\gamma_j}(\phi)$ depend linearly on ϕ .

For continuity we use the Mellin dictionary of Lemma 37 below (whose proof is independent of the results of this section): the coefficient $D_{km}(\phi) = c_{\alpha_k, m-1}(\phi)$ is $1/(m-1)!$ times the coefficient of $(z + \alpha_k)^{-m}$ in the Laurent expansion at $z = -\alpha_k$ of the zeta function $\zeta_\phi(z) = \int_X f^z \phi dx$. Hence, by Cauchy's formula, for a small circle C_r about $-\alpha_k$ containing no other pole,

$$D_{km}(\phi) = \frac{1}{(m-1)!} \cdot \frac{1}{2\pi i} \oint_{C_r} (z + \alpha_k)^{m-1} \zeta_\phi(z) dz.$$

By the iterated Bernstein–Sato continuation (10), there are an integer n (large enough that $\operatorname{Re}(z) + n > 0$ on C_r), a differential operator $\tilde{D}(z)$ of some order L depending polynomially on z , and a polynomial $b(z) = \prod_{i=0}^{n-1} b_f(z+i)$ such that

$$\zeta_\phi(z) = \frac{1}{b(z)} \int_X f(x)^{z+n} \tilde{D}(z)^* \phi(x) dx.$$

On the circle C_r the denominator is bounded below and the integrand is bounded by $C \|\phi\|_{C^L}$ (with C depending on f, r, α_k and $\operatorname{supp} \phi$), so

$$|D_{km}(\phi)| \leq C' \|\phi\|_{C^L}.$$

Thus D_{km} is a linear functional on $C_0^\infty(X)$ bounded in the C^L norm, hence a distribution of order at most L . $\square$

2.8 Appendix: the regular singular ODE

The following lemma was the engine of the first version of this note, in which Theorem 30 was to be proved by an inductive ODE argument. It is *not* used above — the comparison argument of Theorem 30 replaces it — but we record it, in corrected form, since it gives the explicit form of the remainder in the expansion and is the natural companion to (6).

Lemma 35 (Inhomogeneous regular singular ODE). *Let $\alpha > 0$, $\beta > \alpha$, and let $m \geq 1$ be an integer. Suppose $g: (0, \epsilon) \rightarrow \mathbb{R}$ satisfies $g(t) = O(t^{\beta-1} |\log t|^M)$ as $t \rightarrow 0^+$, for some integer $M \geq 0$. If $w: (0, \epsilon) \rightarrow \mathbb{R}$ is a smooth solution of*

$$(-\partial_t t + \alpha)^m w = g,$$

then there exist constants $c_0, \dots, c_{m-1} \in \mathbb{R}$ with

$$w(t) = \sum_{j=0}^{m-1} c_j t^{\alpha-1} (-\log t)^j + O(t^{\beta-1} |\log t|^{M+m-1}) \quad (t \rightarrow 0^+).$$

Proof. Write $L := -\partial_t t + \alpha$, so $Lw = -tw' + (\alpha - 1)w$. Note that L preserves the span $\Sigma_m := \langle t^{\alpha-1} (-\log t)^j : 0 \leq j \leq m-1 \rangle$ and acts there as the nilpotent operator $N: t^{\alpha-1} (-\log t)^j \mapsto j t^{\alpha-1} (-\log t)^{j-1}$. Since L^m is a linear ODE of order m with smooth coefficients and nonvanishing leading coefficient $(-t)^m$ on $(0, \epsilon)$, its solution space there is exactly m -dimensional; as $\Sigma_m \subset \ker L^m$ and $\dim \Sigma_m = m$, we get $\ker L^m = \Sigma_m$. It therefore suffices to exhibit one particular solution of size $O(t^{\beta-1} |\log t|^{M+m-1})$.

Case $m = 1$. Substituting $w = t^{\alpha-1} h$ into $Lw = g$ gives $-t^\alpha h'(t) = g(t)$, i.e. $h'(t) = -t^{-\alpha} g(t)$. Since $\beta > \alpha$, the function $t^{-\alpha} g(t) = O(t^{\beta-\alpha-1} |\log t|^M)$ is integrable at 0, so we may define

$$h(t) := - \int_0^t s^{-\alpha} g(s) ds = O(t^{\beta-\alpha} |\log t|^M),$$

whence $w_p := t^{\alpha-1}h = O(t^{\beta-1}|\log t|^M)$ solves $Lw_p = g$.

Case $m > 1$. Apply the case $m = 1$ repeatedly: define $g_m := g$ and $g_{i-1} :=$ the particular solution of $Lg_{i-1} = g_i$ produced above. An induction shows $g_i = O(t^{\beta-1}|\log t|^{M+(m-i)})$, since each application of the integral operator raises the logarithmic power by at most one; then $w_p := g_0$ satisfies $L^m w_p = g$ and $w_p = O(t^{\beta-1}|\log t|^{M+m-1})$. Adding an element of $\ker L^m = \Sigma_m$ gives the general solution. $\square$

Corollary 36 (Right inverse). *For $\beta > \alpha > 0$, $m \geq 1$ and $M \geq 0$, the operator $(-\partial_t + \alpha)^m$ admits a right inverse R_α on the space of smooth functions on $(0, \epsilon)$ that are $O(t^{\beta-1}|\log t|^M)$, with*

$$(-\partial_t + \alpha)^m (R_\alpha g) = g, \quad R_\alpha g = O(t^{\beta-1}|\log t|^{M+m-1}).$$

For $m = 1$ it is given explicitly by $R_\alpha g(t) = -t^{\alpha-1} \int_0^t s^{-\alpha} g(s) ds$.

Proof. This is the particular solution constructed in the proof of Lemma 35, which is manifestly linear in g . $\square$

3 The zeta function and real log canonical threshold

In this section we address the question: when do the terms of the asymptotic expansion in Theorem 32 vanish? Understanding this requires an alternative derivation of the expansion, which depends on the zeta function of f .

3.1 The zeta function

For $\phi \in C_0^\infty(X)$, define the *zeta function*

$$\zeta_\phi(z) = \int_X f(x)^z \phi(x) dx.$$

This converges absolutely for $\operatorname{Re}(z) > 0$ and defines a holomorphic function there. The analytic continuation to the entire complex plane is provided by the Bernstein–Sato polynomial $b_f(s)$, which satisfies the functional equation

$$D(z, x, \partial_x) f(x)^{z+1} = b_f(z) f(x)^z,$$

where D is a differential operator depending polynomially on z . Integrating by parts:

$$\zeta_\phi(z) = \frac{1}{b_f(z)} \int_X f(x)^{z+1} D^*(z, x, \partial_x) \phi(x) dx. \quad (9)$$

The integral on the right converges absolutely and is holomorphic on the half-plane $\operatorname{Re}(z) > -1$, so (9) continues ζ_ϕ meromorphically to $\operatorname{Re}(z) > -1$, with poles at most at the roots of $b_f(z)$ (those not cancelled by the vanishing of the integral). Iterating n times gives

$$\zeta_\phi(z) = \frac{1}{\prod_{i=0}^{n-1} b_f(z+i)} \int_X f(x)^{z+n} \tilde{D}(z, x, \partial_x)^* \phi(x) dx, \quad \tilde{D}(z) := D(z) D(z+1) \cdots D(z+n-1), \quad (10)$$

which is meromorphic on $\operatorname{Re}(z) > -n$; letting $n \rightarrow \infty$ continues ζ_ϕ to the whole plane. The poles of $\zeta_\phi(z)$ are thus contained in the set of nonpositive integer translates of the roots of $b_f(z)$, and in particular are negative rational numbers. Note for later use (Lemma 34) that (10) exhibits $\zeta_\phi(z)$, on any compact subset of $\{\operatorname{Re}(z) > -n\}$ avoiding the zeros of $\prod_i b_f(z+i)$, as bounded by $C\|\phi\|_{C^L}$, where L is the order of $\tilde{D}$.

3.2 The Mellin dictionary

The zeta function is the Mellin transform of the state density: since v_ϕ is a compactly supported distribution on $[0, \infty)$, locally integrable on $(0, \infty)$, the defining property (4) gives

$$\zeta_\phi(z) = \int_0^\infty t^z v_\phi(t) dt \quad (\operatorname{Re}(z) > 0).$$

The following standard lemma converts the asymptotic expansion of v_ϕ at $t = 0^+$ into the polar data of ζ_ϕ , and back. We stress that its proof is elementary and independent of Section 2.3; this is what allows it to be used in the proof of Lemma 34.

Lemma 37 (Mellin dictionary). *Let $\phi \in C_0^\infty(X)$ and suppose (as guaranteed by Theorem 23) that v_ϕ has asymptotic symbol $\tau(v_\phi) = \sum_{\gamma, j} c_{\gamma j}(\phi) t^{\gamma-1} (-\log t)^j$ with all exponents positive. Then ζ_ϕ is meromorphic on $\mathbb{C}$ with poles contained in $\{-\gamma : \gamma \text{ an exponent}\}$, and for each such γ the Laurent expansion at $z = -\gamma$ is*

$$\zeta_\phi(z) = \sum_{j \geq 0} \frac{j! c_{\gamma j}(\phi)}{(z + \gamma)^{j+1}} + (\text{holomorphic}).$$

In particular ζ_ϕ has a pole at $-\gamma$ of order $j_{\max}(\gamma) + 1$ precisely when γ is an exponent of $\tau(v_\phi)$ with top logarithmic degree $j_{\max}(\gamma)$; and the two families of coefficients determine each other.

Proof. Fix $A > 0$ and write $v_\phi = \sum_{\gamma \leq A, j} c_{\gamma j} t^{\gamma-1} (-\log t)^j \cdot \chi(t) + r_A(t)$, where χ is a cutoff, identically 1 near 0 and supported in $(0, \epsilon)$, and $r_A(t) = O(t^A)$ near 0 (and r_A is compactly supported and locally integrable). Then $\int_0^\infty t^z r_A(t) dt$ is holomorphic on $\operatorname{Re}(z) > -A - 1$. For each model term, the substitution $u = -\log t$ gives

$$\int_0^1 t^{z+\gamma-1} (-\log t)^j dt = \int_0^\infty e^{-(z+\gamma)u} u^j du = \frac{j!}{(z + \gamma)^{j+1}} \quad (\operatorname{Re}(z) > -\gamma),$$

and replacing the cutoff at 1 by χ changes this by an entire function. Summing the finitely many terms with $\gamma \leq A$ and letting $A \rightarrow \infty$ gives the claim. $\square$

Combining Lemma 37 with Theorem 32 gives at once:

Proposition 38. *For every $\phi \in C_0^\infty(X)$, the poles of ζ_ϕ are simple integer translates of nothing more than the V -filtration data: if $-\gamma$ is a pole of ζ_ϕ , then γ is a positive jumping index α_k of the V -filtration on $\iota_+ \mathcal{O}_X$, and the order of the pole is at most m_k , the nilpotency order of $s + \alpha_k$ on $\operatorname{gr}_V^{\alpha_k}$.*

Proof. By Lemma 37, a pole of ζ_ϕ at $-\gamma$ of order θ forces γ to be an exponent of $\tau(v_\phi)$ with top logarithmic degree $\theta - 1$. By Theorem 32, $\gamma = \alpha_k$ for some k and $\theta - 1 \leq m_k - 1$. $\square$

Corollary 39. *We have the inclusion*

$$\bigcup_{\phi \in C_0^\infty(X)} \{-\lambda : \lambda \text{ is a pole of } \zeta_\phi\} \subset \{\text{positive jumping indices of the } V\text{-filtration on } \iota_+ \mathcal{O}_X\}.$$

We will show below that this inclusion is strict.

3.3 The real log canonical threshold

The index of the first nonvanishing term of the expansion of $v(t)$, known as the real log canonical threshold, is especially important in singular learning theory. In the analytic category, it is usually defined as follows:

Definition 40. The *real log canonical threshold* of f is the smallest $\alpha > 0$ such that $|f|^{-\alpha}$ is not locally integrable on X :

$$\text{rlct}(f) = \inf\{\alpha > 0 : |f|^{-\alpha} \notin L^1_{\text{loc}}(X)\}.$$

To connect this definition to the expansion of $v(t)$, let $\pi: X' \rightarrow X$ be a resolution of singularities such that $f \circ \pi$ is locally of the form $u \prod_i x_i^{m_i}$ where u is a nowhere-vanishing real analytic function, and the Jacobian $|\det d\pi|$ is locally $|u'| \prod_i x_i^{a_i}$ for u' nowhere vanishing. Let D'_j denote the irreducible components of the total transform with multiplicities m_j and Jacobian exponents a_j . Define

$$J_{\mathbb{R}} = \{j \in J : D'_j(\mathbb{R}) \neq \emptyset\}.$$

Proposition 41 (Geometric formula for the rlct). *The real log canonical threshold of f is*

$$\text{rlct}(f) = \min_{j \in J_{\mathbb{R}}} \frac{a_j + 1}{m_j}.$$

Proof. By the change of variables formula, $\int |f|^{-\alpha} \phi dx = \sum_{\ell} \int |u|^{-\alpha} \prod_i x_i'^{a_i - \alpha m_i} \Phi_{\ell} dx'$ (via partition of unity on X'), where $\Phi_{\ell} \geq 0$ involves $\phi \circ \pi$ and $|u'|$. Each factor $\int_0^{\epsilon} x_i'^{a_i - \alpha m_i} dx_i'$ converges if and only if $a_i - \alpha m_i > -1$, i.e. $\alpha < (a_i + 1)/m_i$. The integral over the real locus only involves divisors $j \in J_{\mathbb{R}}$ (the others have empty real locus). So $|f|^{-\alpha}$ is locally integrable if and only if $\alpha < (a_j + 1)/m_j$ for all $j \in J_{\mathbb{R}}$, and the threshold is the minimum. $\square$

Using the same resolution of singularities, we can also make a connection to ζ_{ϕ} :

Proposition 42. *Let $\phi \in C_0^{\infty}(X)$ with $\phi \geq 0$, and set $\text{rlct}(f; \phi) := \text{rlct}(f|_U)$ for U a neighbourhood of $\text{supp } \phi$. Then ζ_{ϕ} has no pole in the half-plane $\text{Re}(z) > -\text{rlct}(f; \phi)$, and $-\text{rlct}(f; \phi)$ is the largest pole of ζ_{ϕ} provided $\phi > 0$ at some point of the locus where $|f|^{-\text{rlct}(f; \phi)}$ fails to be integrable. Equivalently, $\text{rlct}(f; \phi)$ is the index of the first nonvanishing term in the asymptotic expansion of $v_{\phi}(t)$. Since $\text{rlct}(f; \phi) \geq \text{rlct}(f)$ for every ϕ , no ζ_{ϕ} has a pole in $\text{Re}(z) > -\text{rlct}(f)$.*

Proof. Using the resolution π as in Proposition 41, the integral $\int |f|^{-\alpha} \phi dx$ transforms to a sum of integrals in normal crossing charts:

$$\int |f|^{-\alpha} \phi dx = \sum_{\ell} \int \prod_i x_i'^{a_i - \alpha m_i} \Phi_{\ell}(x') dx',$$

where $\Phi_{\ell} \geq 0$. Each factor $\int_0^{\epsilon} x_i'^{a_i - \alpha m_i} dx_i'$ converges if and only if $a_i - \alpha m_i > -1$. For $\alpha < \text{rlct}(f)$, all such integrals converge (by definition of the minimum). For $\alpha = \text{rlct}(f)$, there exists $j \in J_{\mathbb{R}}$ with $a_j - \alpha m_j = -1$, and the corresponding integral diverges logarithmically. Choosing ϕ supported near $\pi(D'_{j, \mathbb{R}})$ ensures $\Phi_{\ell} > 0$ on the relevant divisor, so the divergence is not cancelled. $\square$

We can now show that the inclusion of Corollary 39 is strict in general. The argument compares the real V -filtration with its complexification, and it must be run *locally at a real point*: the complex log canonical threshold of $f_{\mathbb{C}}$ may be computed at points of $f_{\mathbb{C}}^{-1}(0)$ lying away from the real locus, where no comparison with the real theory is available.

Proposition 43. *Suppose $\text{rlct}(f) > 1$. Then the V -filtration on $\iota_+ \mathcal{O}_X$ has a positive jumping index $\alpha \leq 1$ such that $-\alpha$ is not a pole of ζ_{ϕ} for any $\phi \in C_0^{\infty}(X)$. In particular the inclusion of Corollary 39 is strict.*

Proof. Fix a point $x_0 \in f^{-1}(0)$ and let $b_{f,x_0}(s)$ denote the local Bernstein–Sato polynomial at x_0 ; the jumping indices of the V -filtration are local invariants, in the sense that $\text{gr}_V^\gamma \neq 0$ if and only if the stalk $(\text{gr}_V^\gamma)_x$ is nonzero for some x .

Step 1: the local jumping indices at x_0 are those of the complexification. The ring of germs at x_0 of real analytic functions has the ring of germs at $x_0 \in X_{\mathbb{C}}$ of holomorphic functions as a faithfully flat extension (Remark 6). Extending scalars therefore carries the germ at x_0 of $\iota_+ \mathcal{O}_X$ to that of $\iota_+ \mathcal{O}_{X_{\mathbb{C}}}$, together with the actions of t and s , and carries the filtration $V_{x_0}^\bullet$ to a filtration satisfying axioms (i)–(iv) over $\mathbb{C}$. By uniqueness of the V -filtration, the latter is the complex V -filtration at x_0 . Hence $\text{gr}_{V,x_0}^\gamma \otimes_{\mathbb{R}} \mathbb{C} \cong \text{gr}_{V_{\mathbb{C}},x_0}^\gamma$, and the local jumping indices (with their nilpotency orders) coincide.

Step 2: $\alpha := \text{lct}_{x_0}(f_{\mathbb{C}}) \leq 1$ is a local jumping index at x_0 . By [4, Theorem 2.7.2], $\text{lct}_{x_0}(f_{\mathbb{C}})$ is the negative of the largest root of $b_{f_{\mathbb{C}},x_0}(s)$, and it is the smallest positive jumping index of the complex V -filtration at x_0 . By Step 1 it is therefore a positive jumping index of the real V -filtration. Moreover $\alpha \leq 1$, because $(s+1)$ always divides $b_f(s)$ (for f vanishing at x_0), so -1 is a root and the largest root is ≥ -1 .

Step 3: α is not a pole. By Proposition 42, for every ϕ the largest pole of ζ_ϕ is $-\text{rlct}(f|_{\text{supp } \phi}) \leq -\text{rlct}(f) < -1 \leq -\alpha$. So ζ_ϕ has no pole at $-\alpha$, for any ϕ . $\square$

Note that $\text{lct}(f_{\mathbb{C}}) \leq \text{rlct}(f)$ always, and the content of Proposition 43 is precisely that the inequality is strict whenever $\text{rlct}(f) > 1$, since $\text{lct}(f_{\mathbb{C}}) \leq 1$ always. For more on the difference between the lct and the rlct , see [5].

Example 44. Let $f(x, y, z) = x^2 + y^2 + z^2$ on $X = \mathbb{R}^3$. In Example 4 we computed $b_f(s) = (s+1)(s+\frac{3}{2})$, with $(\partial_x^2 + \partial_y^2 + \partial_z^2)f^{s+1} = 4(s+1)(s+\frac{3}{2})f^s$; thus $\text{lct}(f_{\mathbb{C}}) = 1$, and 1 is a jumping index of the V -filtration.

However, ζ_ϕ has no pole at $z = -1$. Indeed, (9) reads $\zeta_\phi(z) = (4b_f(z))^{-1} \int_{\mathbb{R}^3} f^{z+1} \Delta \phi \, dx$ (the operator $D = \frac{1}{4}\Delta$ being independent of z , and self-adjoint), so

$$\lim_{z \rightarrow -1} (z+1) \zeta_\phi(z) = \frac{1}{4(-1+\frac{3}{2})} \int_{\mathbb{R}^3} f^0 \Delta \phi \, dx = \frac{1}{2} \int_{\mathbb{R}^3} \Delta \phi \, dx = 0$$

for every $\phi \in C_0^\infty(\mathbb{R}^3)$. So the simple pole of $b_f(z)^{-1}$ at $z = -1$ is always cancelled — not because the operator degenerates, but because the integral of a Laplacian of a compactly supported function vanishes. (This is the elementary shadow of the fact that -1 is a root of b_f but not of the reduced Bernstein–Sato polynomial $\tilde{b}_f(s) = b_f(s)/(s+1) = s + \frac{3}{2}$.)

Finally, $\text{rlct}(f) = \frac{3}{2}$: blowing up the origin gives $f \circ \pi = r^2 u$ with u a unit and $|\det d\pi| = r^2$, so Proposition 41 yields $\text{rlct}(f) = (2+1)/2 = \frac{3}{2} > 1$. This is the situation of Proposition 43 with $\alpha = 1$.

References

- [1] N. Budur, M. Saito, *Multiplier ideals, V-filtration, and spectrum*, Math. Ann. **327** (2003), 257–270.
- [2] J.-E. Björk, *Analytic $\mathcal{D}$ -Modules and Applications*, Mathematics and its Applications, vol. 247, Kluwer Academic Publishers, 1993.
- [3] H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. **79** (1964), 109–326.
- [4] M. Popa, *$\mathcal{D}$ -modules in birational geometry*, Lecture notes.
- [5] M. Saito, *On real log canonical thresholds*, preprint, arXiv:0707.2308, 2007.

- [6] S. Watanabe, *Algebraic Geometry and Statistical Learning Theory*, Cambridge Monographs on Applied and Computational Mathematics, Cambridge University Press, 2009.
- [7] P. Jeanquartier, *Développement asymptotique de la distribution de Dirac attachée à une fonction analytique*, C. R. Acad. Sci. Paris Sér. A **271** (1970), 1159–1161.
- [8] B. Malgrange, *Intégrales asymptotiques et monodromie*, Ann. Sci. École Norm. Sup. (4) **7** (1974), 405–430.
- [9] D. Barlet, *Développement asymptotique des fonctions obtenues par intégration sur les fibres*, Invent. Math. **68** (1982), 129–174.
- [10] V. I. Arnold, S. M. Gusein-Zade, A. N. Varchenko, *Singularities of Differentiable Maps, Volume II: Monodromy and Asymptotics of Integrals*, Birkhäuser, 1988.
- [11] M. F. Atiyah, *Resolution of singularities and division of distributions*, Comm. Pure Appl. Math. **23** (1970), 145–150.